

# Mmesoma Nwachukwu

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## PROFILE

AI/ML Engineer 2026 graduate with a First Class from Coventry University. Built machine learning systems at various companies and deployed production models in a startup environment. Experienced in computer vision and LLM systems, including RAG pipelines and scalable backend infrastructure. Recipient of 5 scholarships and awards across a 400-person cohort and nationwide competitions.

## EDUCATION

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|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>BSc Computer Science</b> <a href="#">Coventry University</a>                                                                           | <i>Expected May 2026</i> |
| <i>Predicted First Class Honours</i> Relevant modules: AI, Data Science, Advanced Algorithms, Software Engineering, Theory of Computation |                          |

## SCHOLARSHIPS AND AWARDS

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|--------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Year in Industry Student of the Year</b><br>1 of 10 selected from 400 placement students nationwide | <i>National Highways, 2025</i>   |
| <b>AWS Machine Learning Scholarship</b><br>Top 2% of 50,000+ DeepRacer applicants                      | <i>Amazon Web Services, 2023</i> |
| <b>Santander Universities Scholarship</b><br>1 of 100 recipients selected nationwide                   | <i>Santander Bank, 2023</i>      |
| <b>Intern of the Year</b>                                                                              | <i>Paradigm Finance, 2021</i>    |
| <b>Best Foundation Student</b><br>Top student out of 50 in the cohort                                  | <i>16 Plus Schools, 2021</i>     |

## PROFESSIONAL EXPERIENCE

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| <b>Software Engineer (Volunteering)</b> <a href="#">Samaritan's Purse</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <i>July 2026 - Present</i> |
| <ul style="list-style-type: none"><li>Designing and building a digital ordering and stock management system for a £10,000 charity procurement project supplying discounted shoebox gifts to 50 community groups</li><li>Building the entire system solo, including an online order form, automated payment and confirmation flow, and a live stock dashboard with low stock alerts.</li><li>Replacing a manual, spreadsheet-based process with an automated pipeline to cut admin time and reduce order errors</li></ul>                                                                                         |                            |
| <b>AI Engineer (Contractor)</b> <a href="#">Projekt Rising</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <i>Dec 2025 - Mar 2026</i> |
| <ul style="list-style-type: none"><li>Analysed client workflows across 3 SMEs to identify automation gaps, then designed and delivered AI integrations that automated 3 business processes, cutting roughly 60 hours of manual work per month across the client operations.</li><li>Built and monitored production APIs connecting AI models to client data sources, tracking accuracy, latency, and user outcomes across deployments.</li></ul>                                                                                                                                                                 |                            |
| <b>Software Engineer Intern</b> <a href="#">SalesAPE.ai</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Dec 2025 - Mar 2026</i> |
| <ul style="list-style-type: none"><li>Designed and built 3 MVP AI prototypes as part of a fast-moving hacksquad, delivering demos to leadership that led to 2 of the 3 projects getting greenlit for further development.</li><li>Designed evaluation pipelines to measure model reliability and document experiment results, moving prototypes closer to production quality.</li></ul>                                                                                                                                                                                                                          |                            |
| <b>Digital Services Year in Industry</b> <a href="#">National Highways</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <i>Jul 2024 - Jul 2025</i> |
| <ul style="list-style-type: none"><li>Built a weather classification and removal model using CNNs achieving 98% classification accuracy on a dataset of 10,000 CCTV images, improving the reliability of road monitoring systems.</li><li>Developed a RAG knowledge graph over thousands of interconnected engineering documents, enabling structured insight extraction to support infrastructure asset management.</li><li>Built a computer vision segmentation model on satellite imagery to identify and classify roadside habitats, improving environmental workflows in infrastructure planning.</li></ul> |                            |

**Founder** FaceRegg*May 2023 - Aug 2023*

- Founded and built an image verification SaaS using CNNs for facial and image matching, acquiring 2 paying clients and generating £5k in revenue through automated identity validation.
- Designed model training and inference pipelines using TensorFlow and OpenCV, optimising accuracy and response latency for real-world inputs.

**Research Intern** Imperial College London*Jun 2023 - Jul 2023*

- Collaborated in a cross-functional team prototyping data-driven technologies addressing climate resilience, working with researchers and industry mentors to assess impact and feasibility.

**Software Engineering Intern** Paradigm Finance*Sept 2021 - Dec 2021*

- Built internal tools for credit risk analysis, generating client portfolio reports and maintaining web applications while adhering to financial data security standards.

**PERSONAL PROJECTS**

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**Sentinel1**

- Built a dual-branch ResNet50 system identifying vehicle make, model, and year from images in 46ms, achieving 81.71% hierarchical accuracy using ordinal regression, Kendall uncertainty weighting, and temperature scaling calibration.
- Deployed a Flask REST API with real-time Euro NCAP database lookup via fuzzy matching, producing court-admissible forensic JSON outputs for accident reconstruction pipelines.

**ExcelPoint**

- Built a full-stack RAG learning platform using Django, React/TypeScript, PostgreSQL, and LangChain with semantic search and context-aware chat over uploaded educational documents using vector embeddings and sentence transformers.
- Architected async processing pipelines with Celery and Redis for document analysis, quiz generation, and adaptive learning workflows, integrated with AWS S3 and JWT-authenticated APIs.

**Payment Authorisation Simulator**

- Built a CLI-first payment authorization simulator modeling the full merchant to issuer flow, covering issuer behavior profiles, retry logic, acquirer routing, 3DS decisioning, and delayed chargeback labels.
- Developed issuer-aware retry and routing strategies that lifted simulated authorization rate from 82.5% to 84.3%, cut wasted retries from 8.1% to 4.2%, and delivered 156 net additional approvals worth an estimated £26,500 net benefit.

**SKILLS**

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**ML / AI:** PyTorch, TensorFlow, LangChain, Hugging Face, OpenCV, Scikit-learn, RAG, Knowledge Graphs, Reinforcement Learning

**Backend:** Python, Django, Flask, Node.js, PostgreSQL, MongoDB, Redis, Celery, REST APIs, SQL

**Frontend:** React, TypeScript, JavaScript, HTML/CSS

**Cloud / Infra:** AWS S3, Databricks, Docker, Git, Jupyter, VS Code

**Languages:** Python, C++, JavaScript, TypeScript, R, SQL